

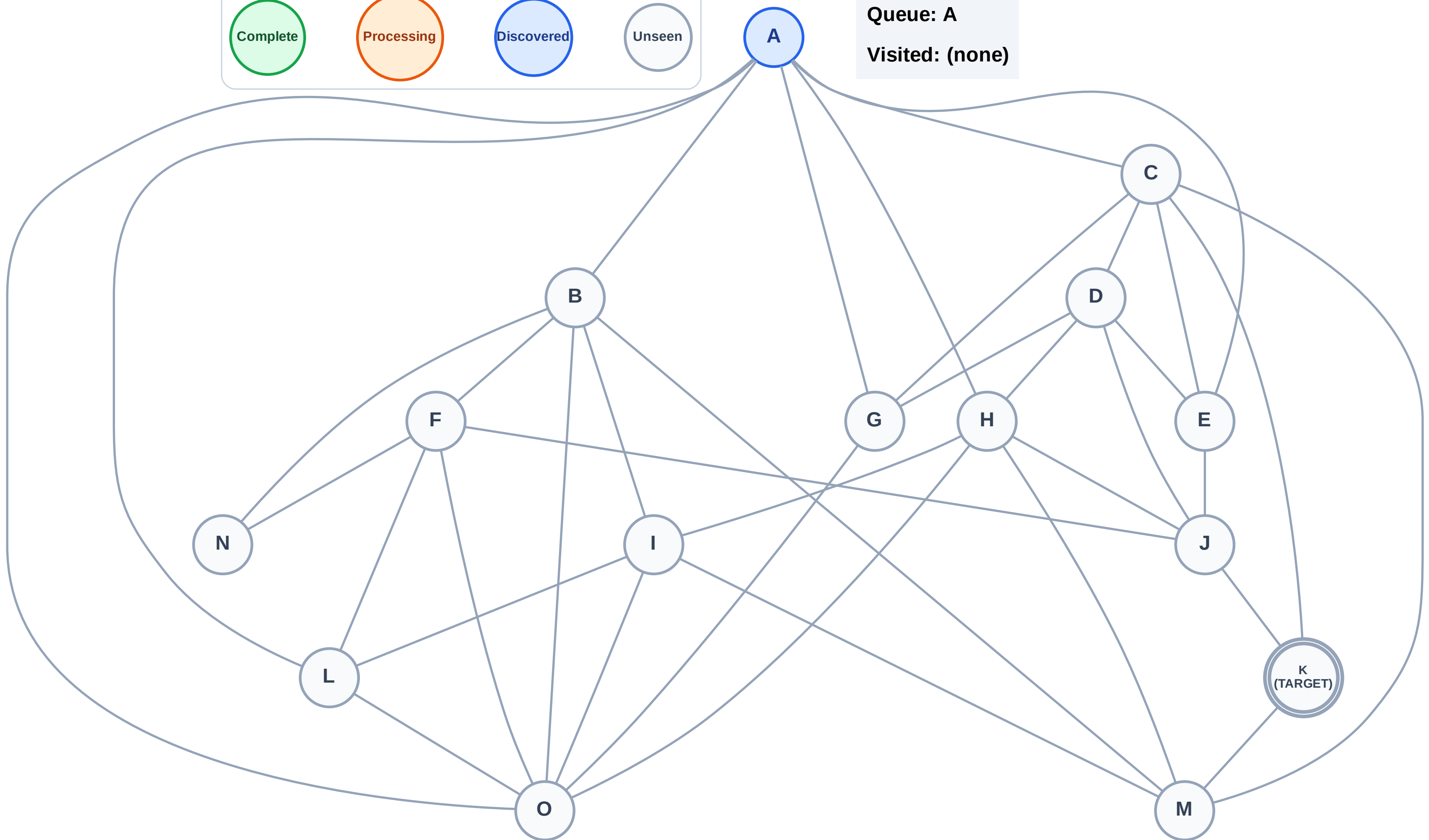
BFS Traversal

Step 1: Start at A (Seeking target node K (Eager Short-Circuit))

Legend



Queue: A
Visited: (none)



BFS Traversal

Step 2: Process node A

Legend

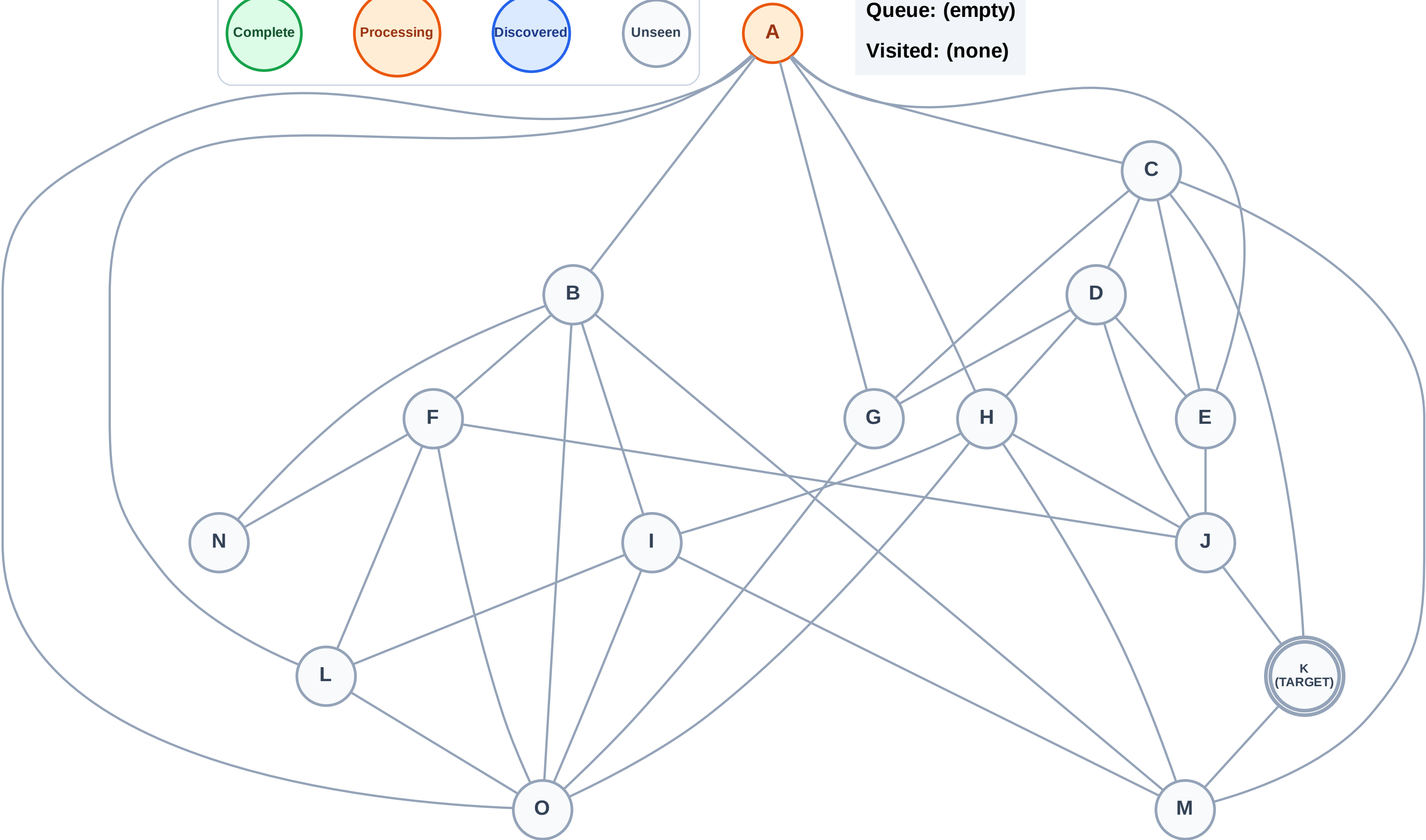
Complete

Processing

Discovered

Unseen

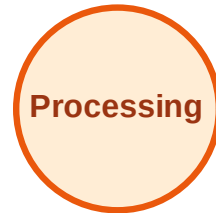
Queue: (empty)
Visited: (none)



BFS Traversal

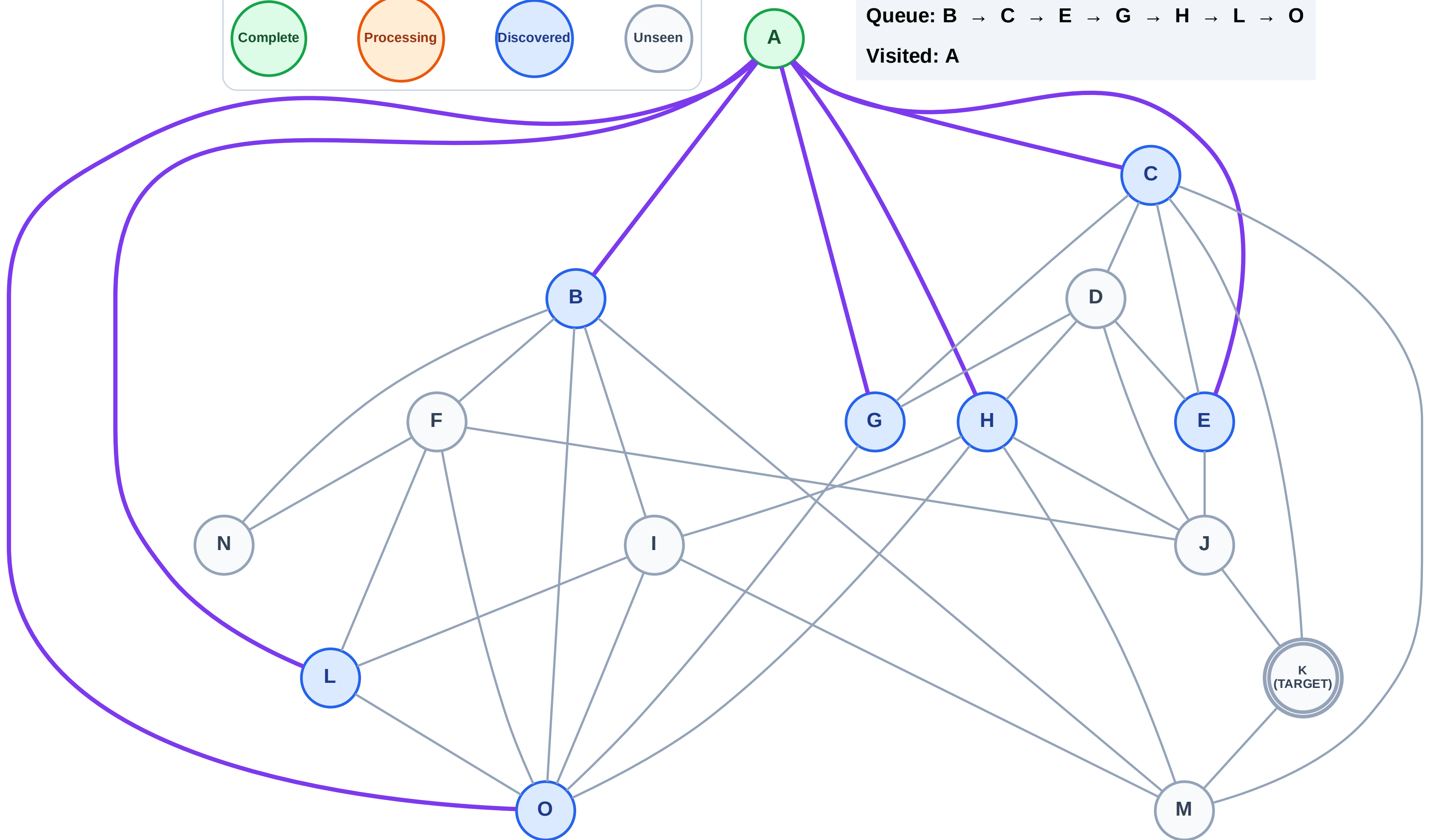
Step 3: Discover B, C, E, G, H, L, O from A

Legend



Queue: B → C → E → G → H → L → O

Visited: A



BFS Traversal

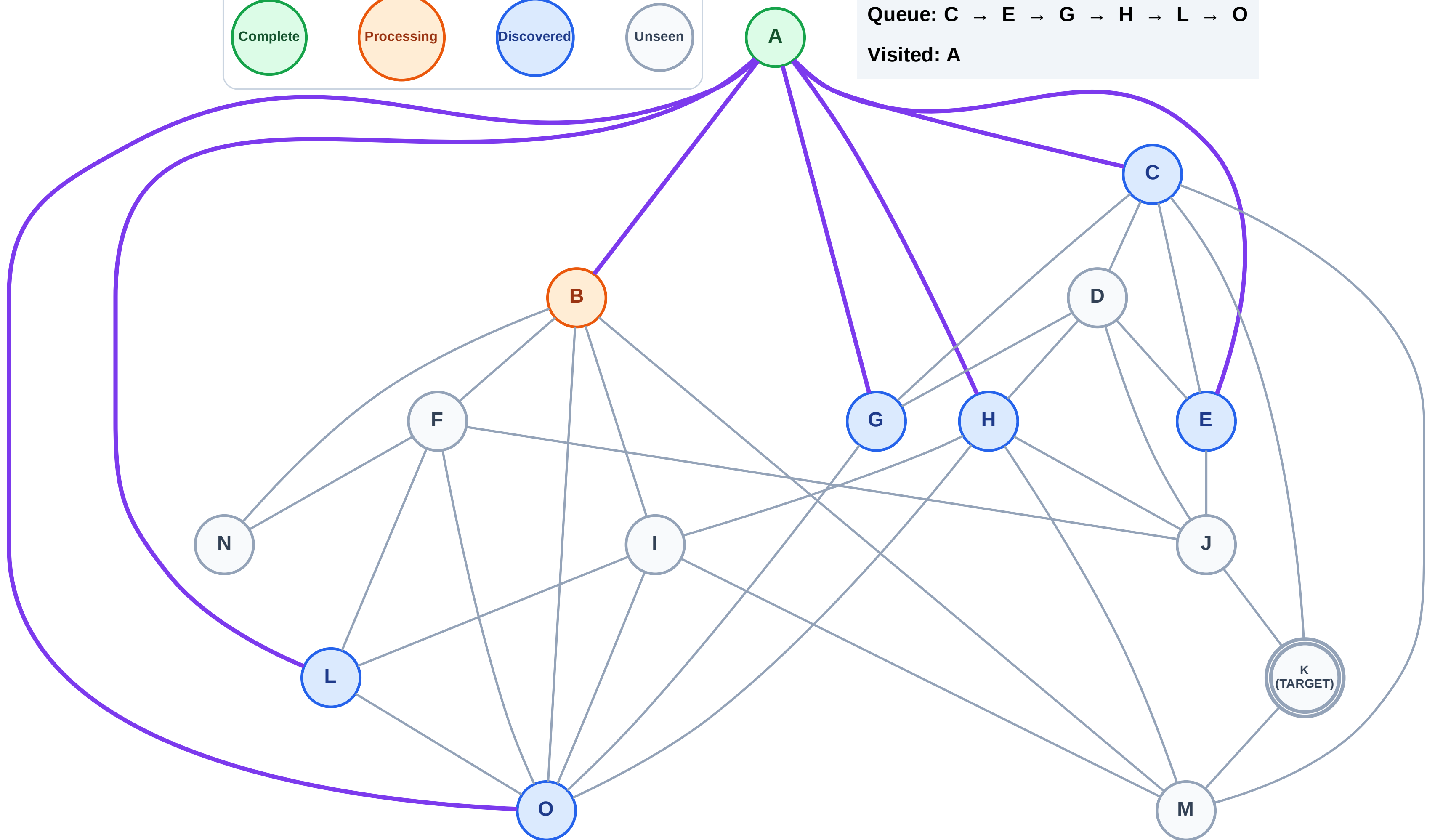
Step 4: Process node B

Legend



Queue: C → E → G → H → L → O

Visited: A



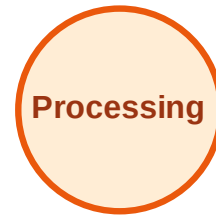
BFS Traversal

Step 5: Discover F, I, M, N from B

Legend



Complete



Processing



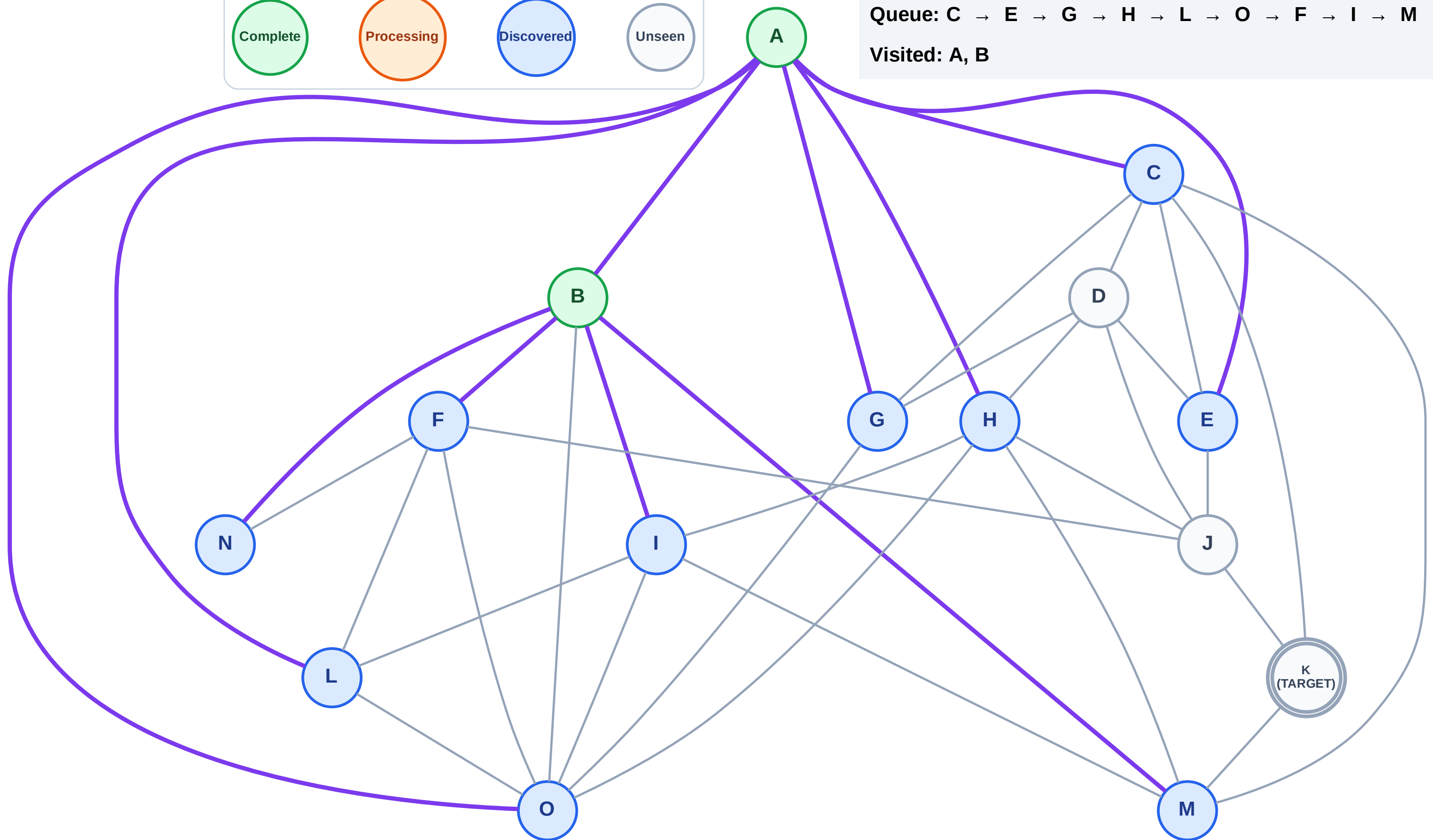
Discovered



Unseen

Queue: C → E → G → H → L → O → F → I → M → N

Visited: A, B



BFS Traversal

Step 6: Process node C

Legend

Complete

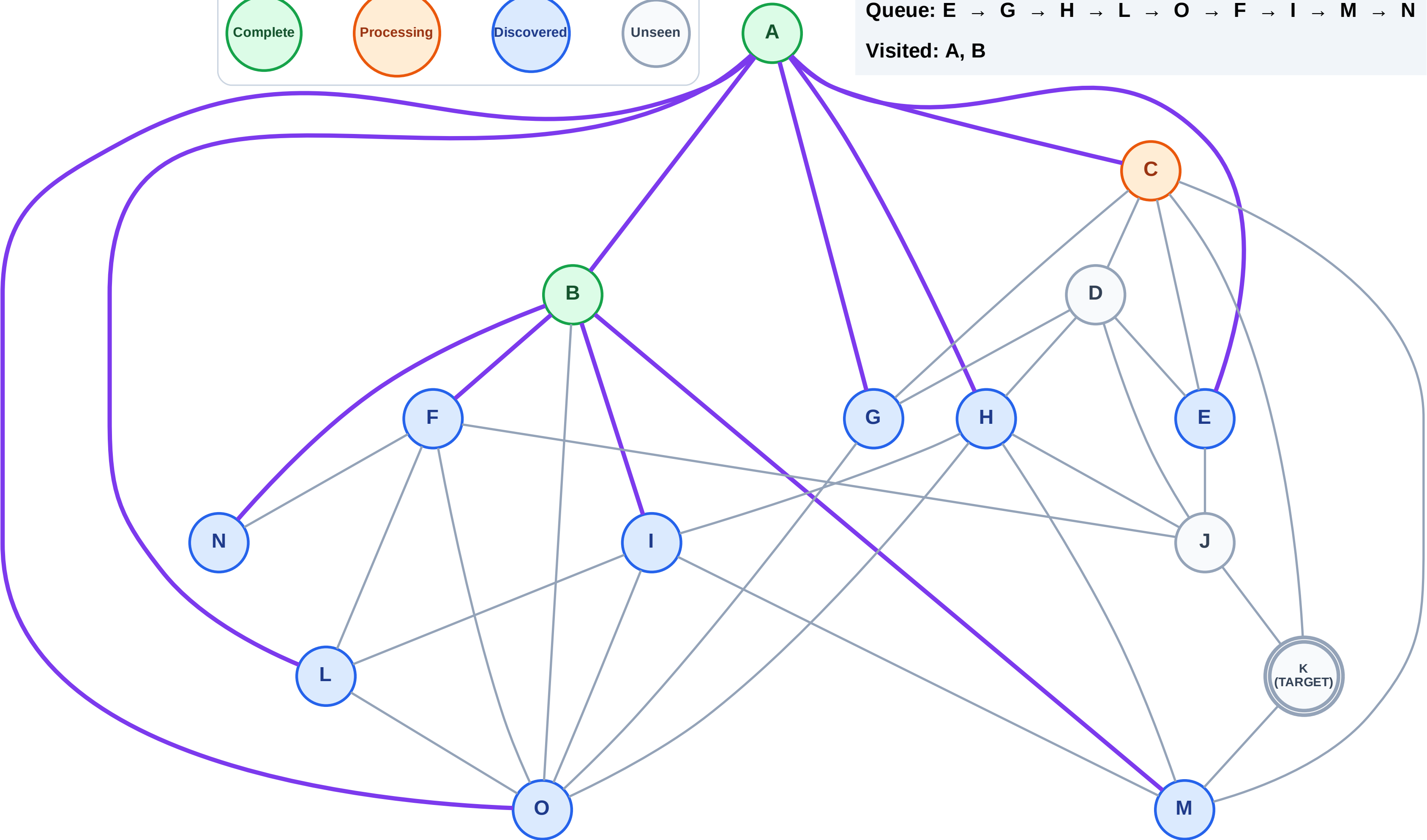
Processing

Discovered

Unseen

Queue: E → G → H → L → O → F → I → M → N

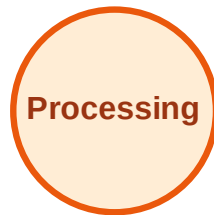
Visited: A, B



BFS Traversal

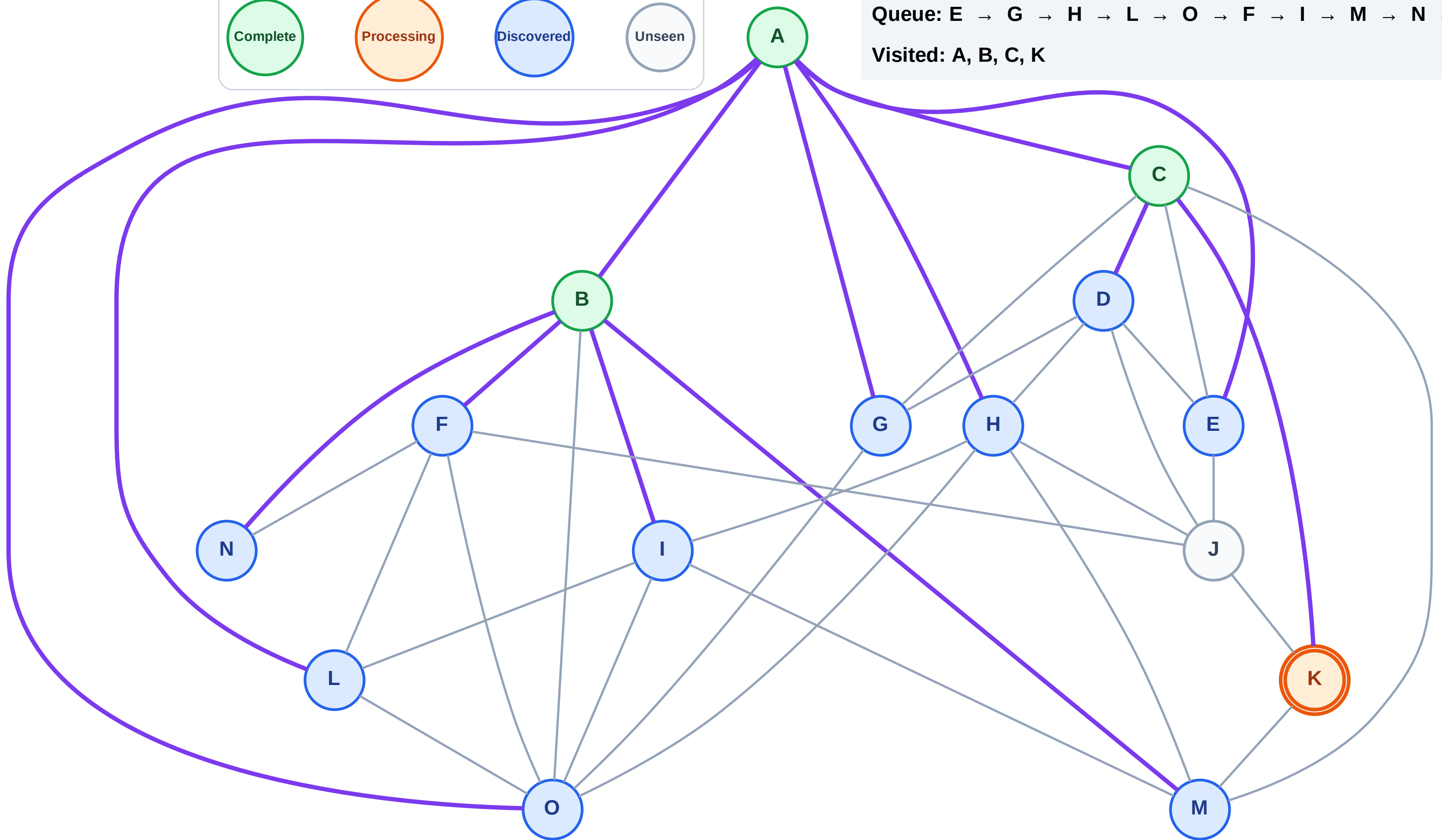
Step 7: TARGET NODE K FOUND (SHORT-CIRCUITED)! Path (2 hops): A → C → K

Legend



Queue: E → G → H → L → O → F → I → M → N → D

Visited: A, B, C, K



DFS Traversal

Step 1: Start at A (Seeking target node K (Eager Short-Circuit))

Legend

Complete

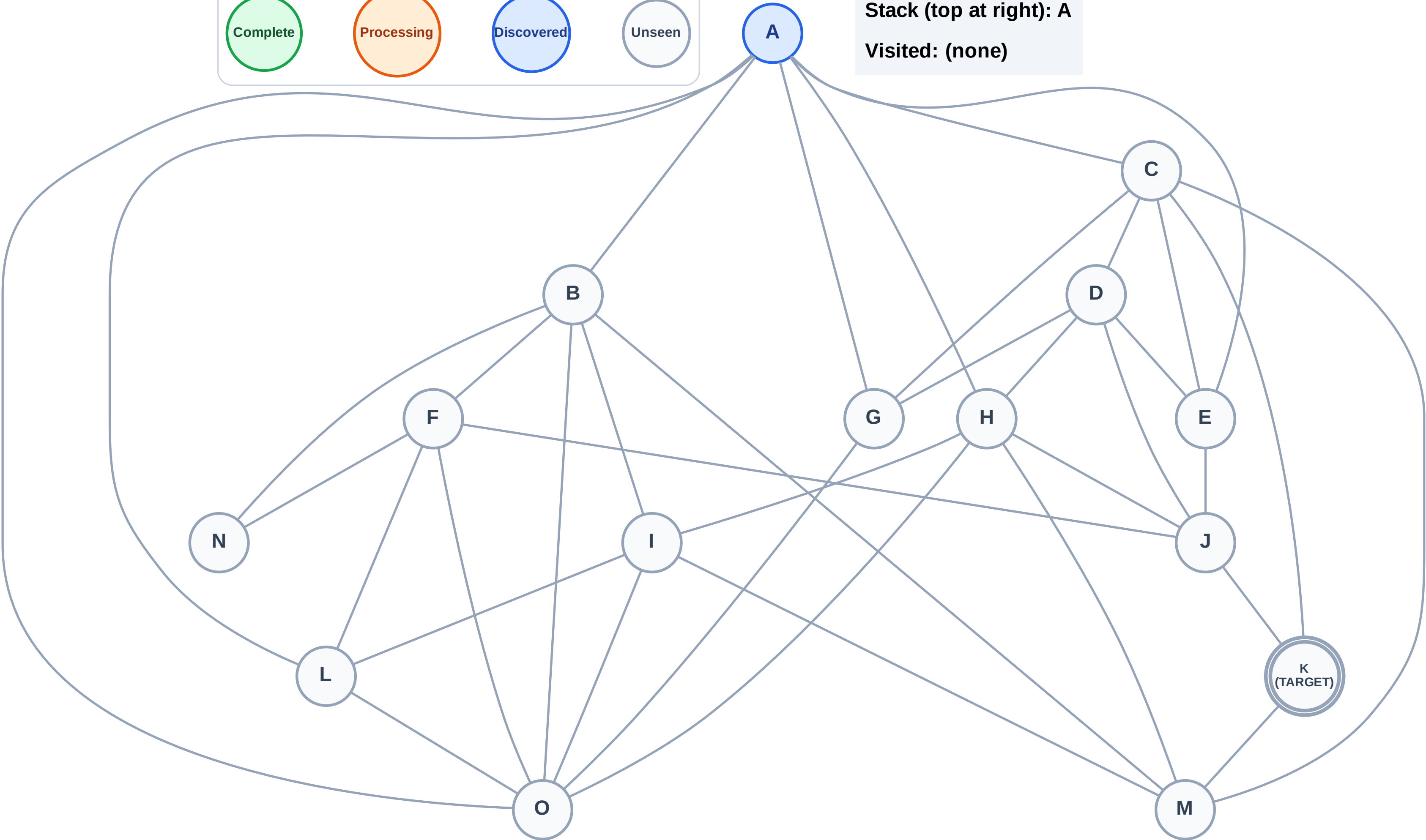
Processing

Discovered

Unseen

Stack (top at right): A

Visited: (none)



DFS Traversal

Step 2: Process node A

Legend

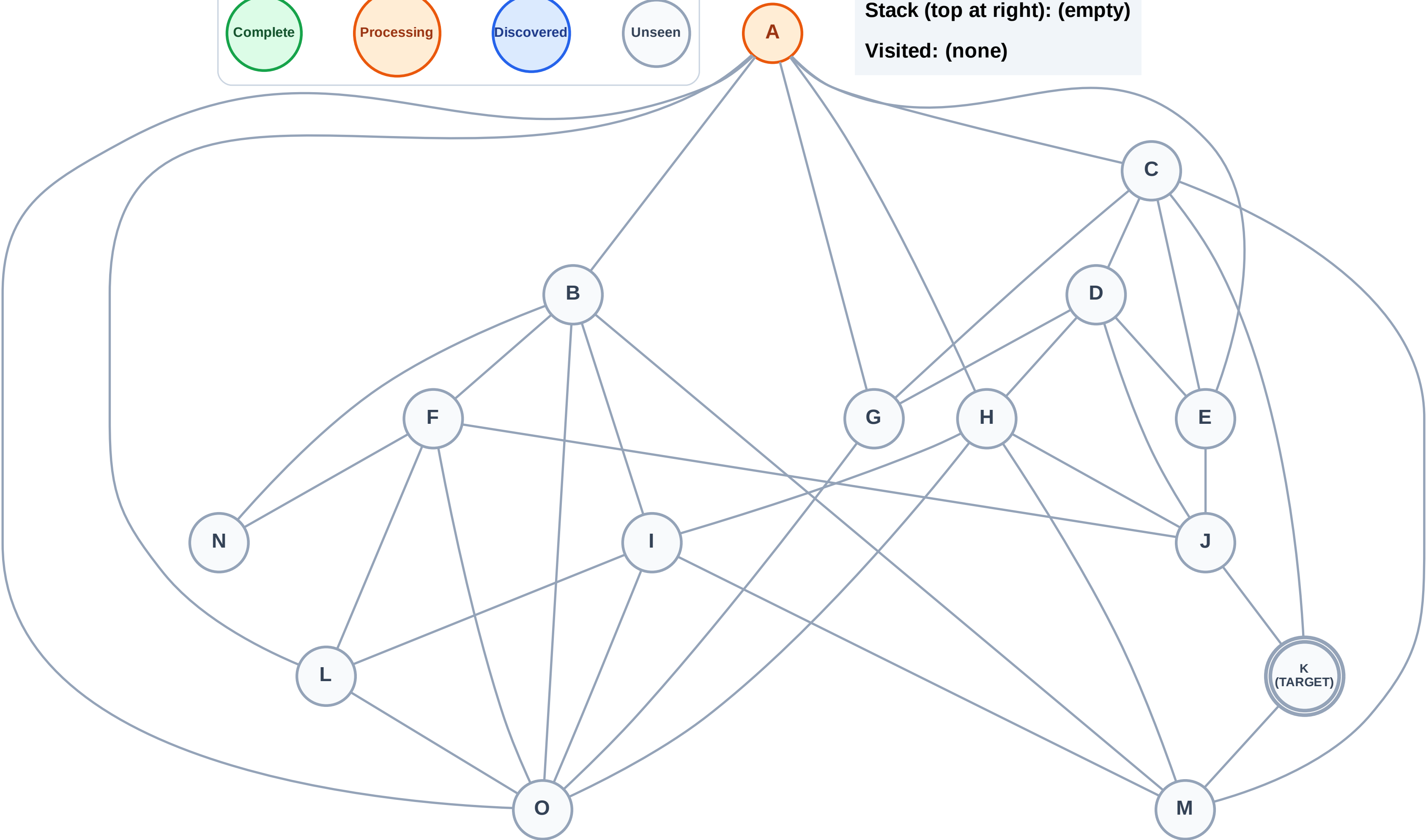
Complete

Processing

Discovered

Unseen

Stack (top at right): (empty)
Visited: (none)



DFS Traversal

Step 3: Discover O, L, H, G, E, C, B from A

Legend

Complete

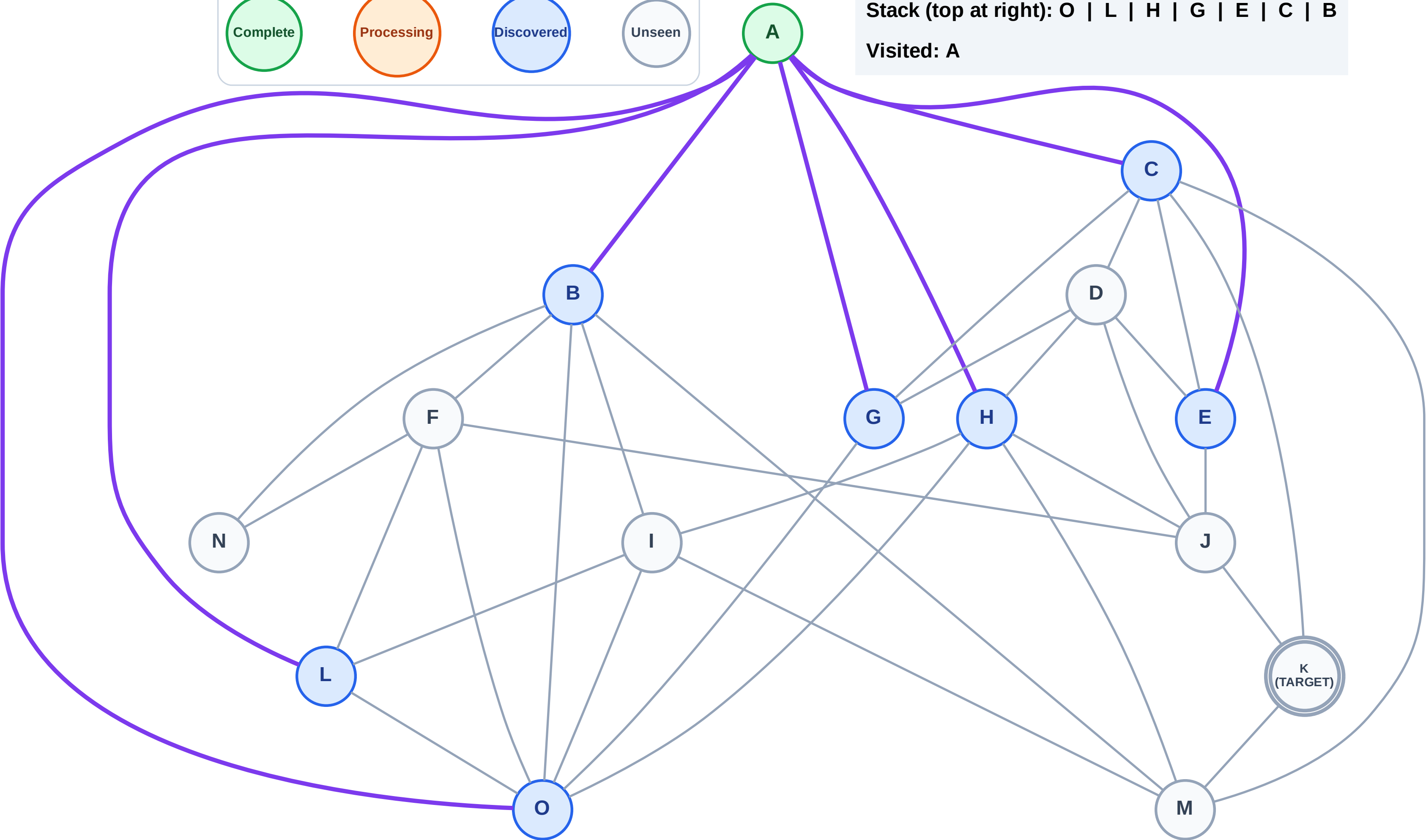
Processing

Discovered

Unseen

Stack (top at right): O | L | H | G | E | C | B

Visited: A



DFS Traversal

Step 4: Process node B

Legend

Complete

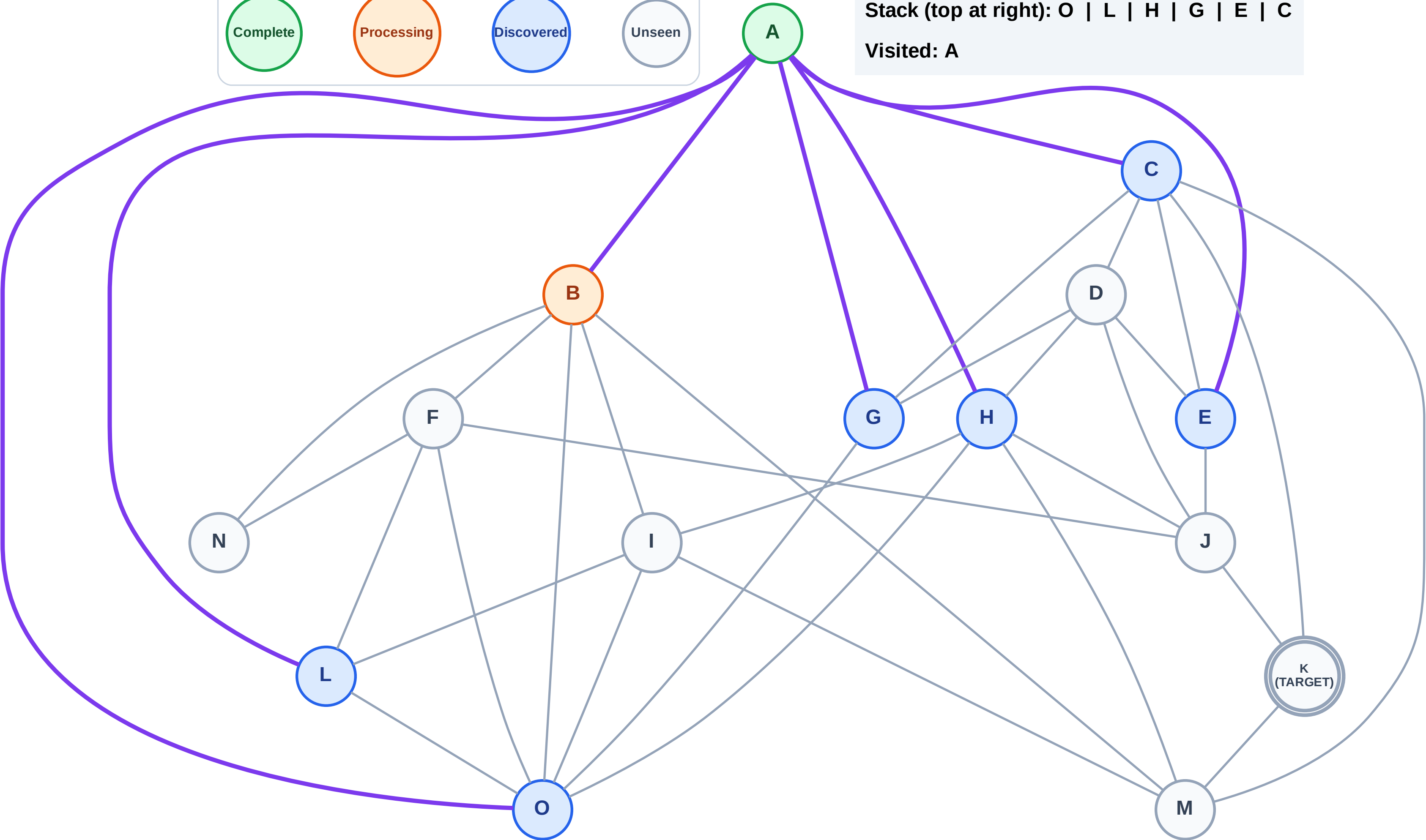
Processing

Discovered

Unseen

Stack (top at right): O | L | H | G | E | C

Visited: A



DFS Traversal

Step 5: Discover N, M, I, F from B

Legend



Complete



Processing



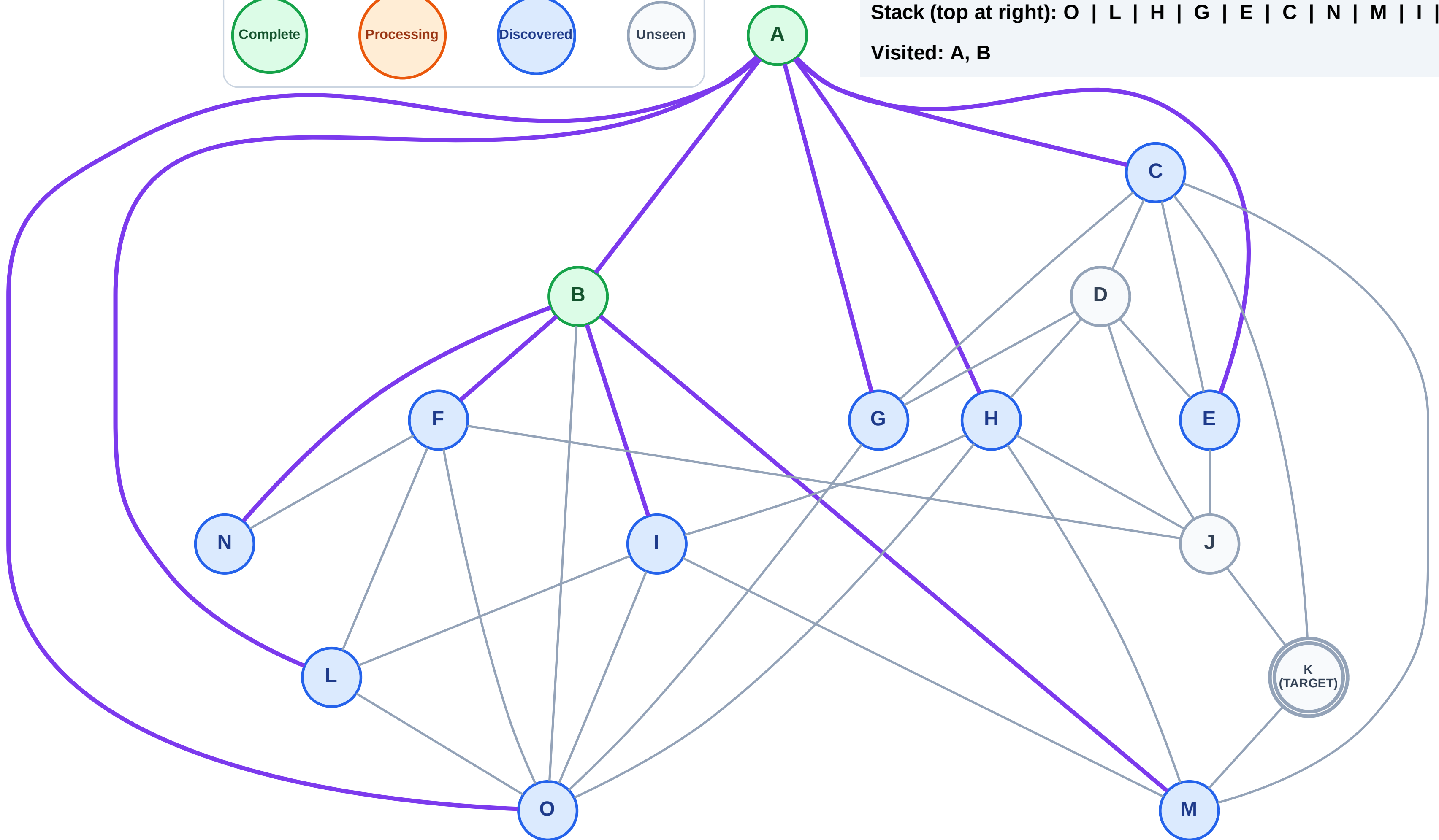
Discovered



Unseen

Stack (top at right): O | L | H | G | E | C | N | M | I | F

Visited: A, B



DFS Traversal

Step 6: Process node F

Legend

Complete

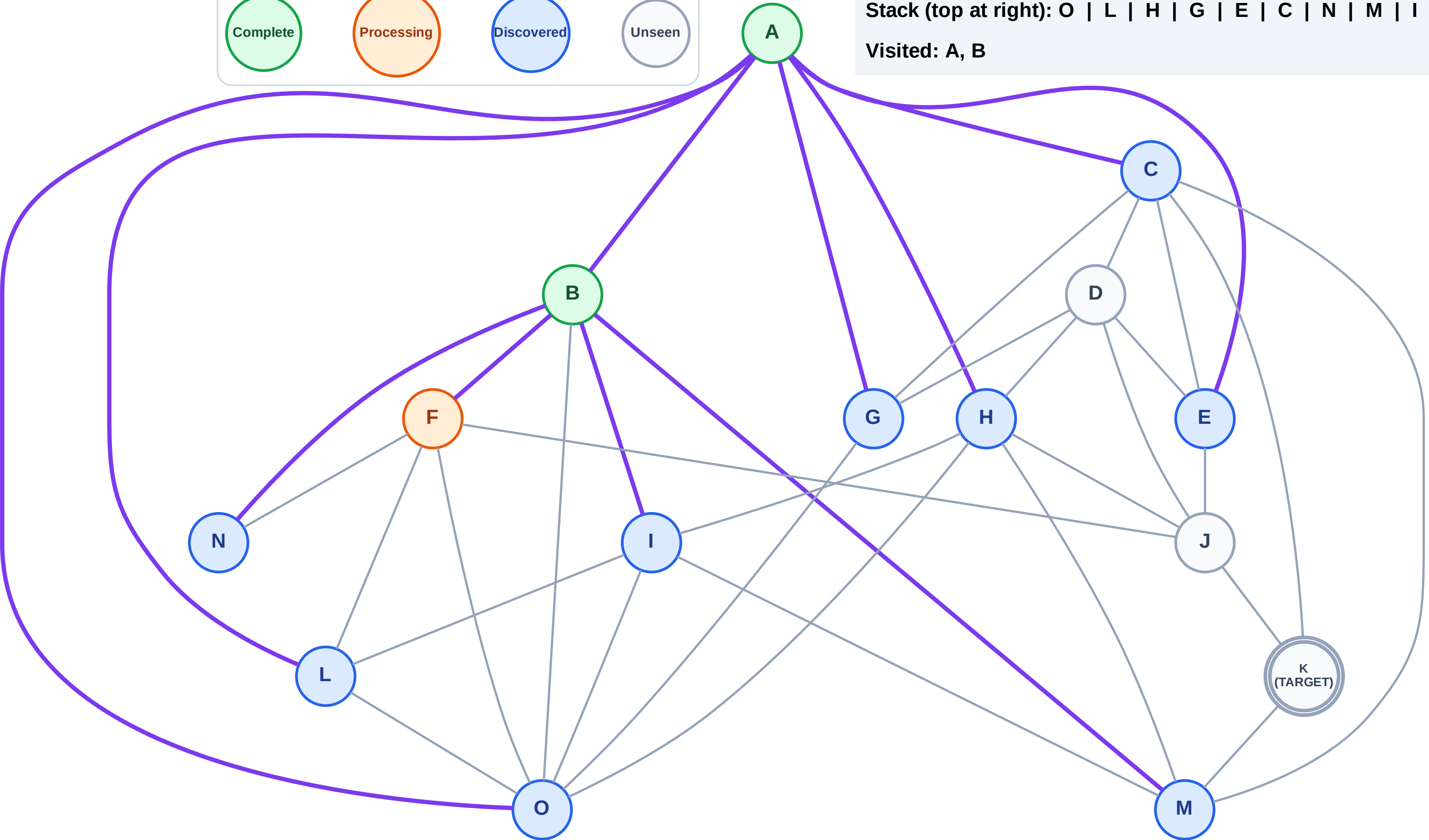
Processing

Discovered

Unseen

Stack (top at right): O | L | H | G | E | C | N | M | I

Visited: A, B



DFS Traversal

Step 7: Discover J from F

Legend

Complete

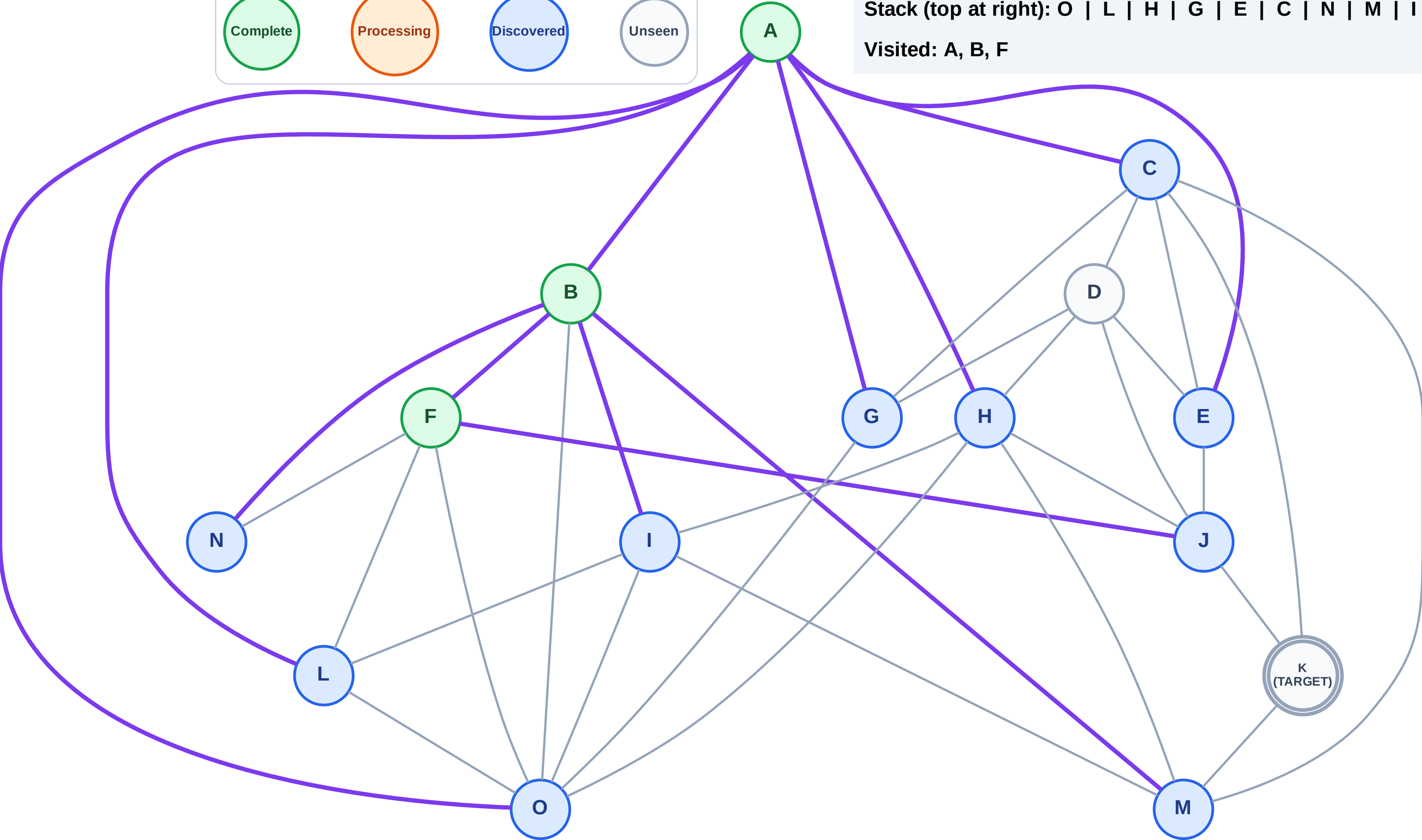
Processing

Discovered

Unseen

Stack (top at right): O | L | H | G | E | C | N | M | I | J

Visited: A, B, F



DFS Traversal

Step 8: Process node J

Legend

Complete

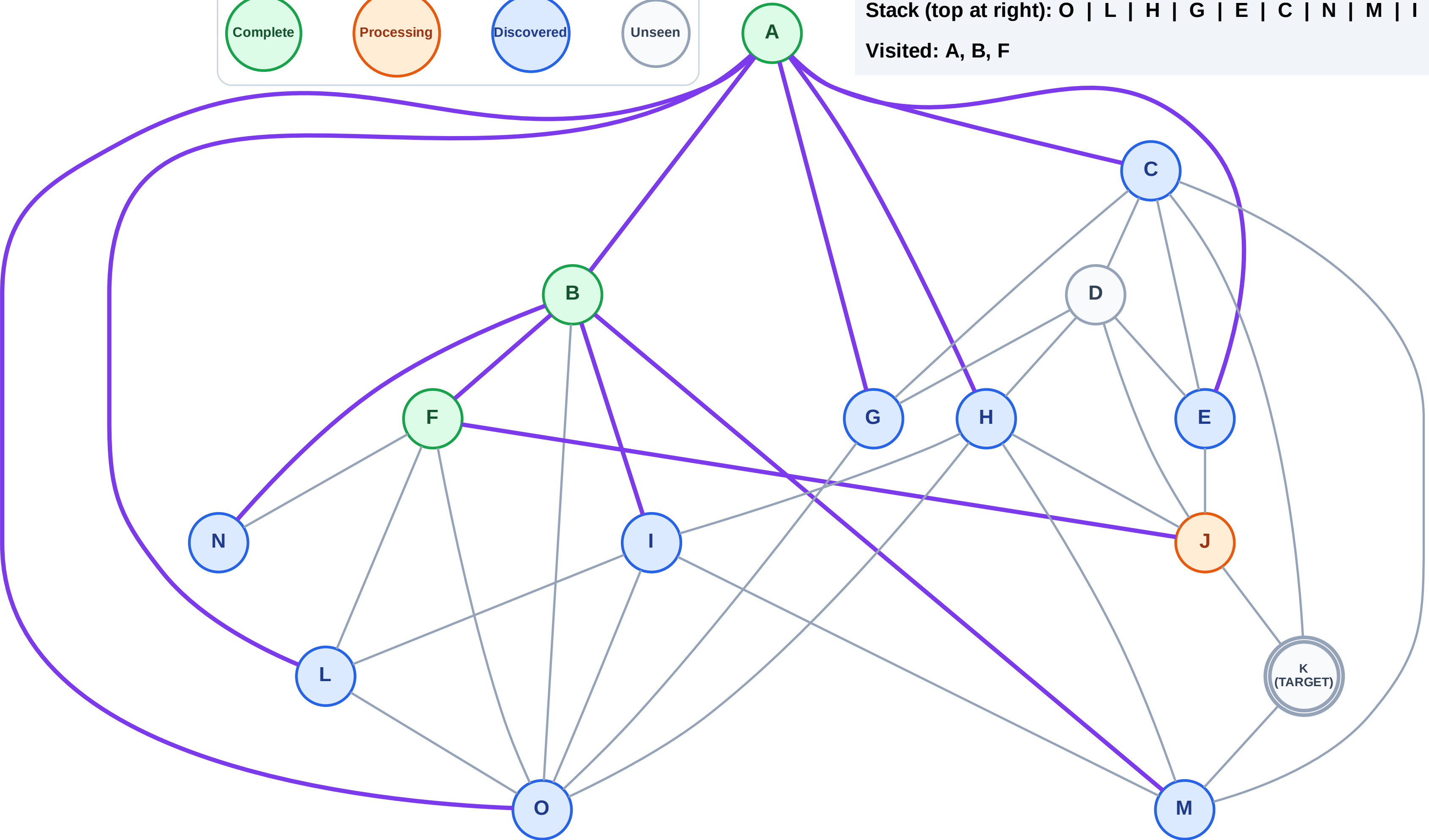
Processing

Discovered

Unseen

Stack (top at right): O | L | H | G | E | C | N | M | I

Visited: A, B, F



DFS Traversal

Step 9: TARGET NODE K FOUND (SHORT-CIRCUITED)! Path (4 hops): A → B → F → J → K

Legend

Complete

Processing

Discovered

Unseen

Stack (top at right): O | L | H | G | E | C | N | M | I

Visited: A, B, F, J, K

